INITIAL STRUCTURAL INTEGRITY ASSESSMENT REPORT (SIAR)

Factory Name:	**ULTIMATE FASHIONS LIMITED**
Address:	**South Gouripur, Ashulia, Savar Ashulia, Savar Dhaka Bangladesh**
Assessor:	**Shaheedullah**
Date:	**16 May 2016**

Introduction to the Report

The following report contains a site profile and summary of non-conformities identified during an onsite assessment commissioned by the Alliance for Bangladesh Worker Safety (Alliance) and conducted by a third-party Qualified Assessment Firm (QAF). The assessment was conducted against the Alliance for Bangladesh Worker Safety Assessment Protocols (APs) and Fire Safety and Structural Integrity Standard, which is harmonized with the factory assessment guidelines developed by Bangladesh University of Engineering and Technology (BUET) for the Bangladesh National Tripartite Plan of Action (NTPA). The goal of the Alliance process is to provide clear and practical technical requirements by which Bangladeshi Ready Made Garment (RMG) Factories producing for Alliance members may be consistently and fairly evaluated for fire, structural, and electrical safety in a non-duplicative manner. Each assessment will prompt action plans that will be used by RMG factories to systematically and sustainably improve safety conditions for garment workers. Beyond tracking and reporting on action steps taken in a transparent manner, the Alliance organization and its members will seek to further support factory improvements through technical assistance, training, implementation support for functional Worker Committees, and in some cases financial assistance and wage support for workers if factories are closed for remediation.

The contents of the report do not constitute a guarantee of compliance with the applicable laws, the Alliance Standard or the absolute or continued safety against fire, electrical and/or structural integrity issues that may lead to injury or loss of life. The report is designed to provide a non-exhaustive summary of risk issues, based on a limited sampling and duration of time onsite by the named QAF. Neither the QAF nor the Alliance can certify or guarantee the quality, outcome, or effectiveness of actions taken in response to the report.

For more information and report feedback please go to: www.bangladeshworkersafety.org.



GENERAL INFORMATION

General Information	
Factory Name:	ULTIMATE FASHIONS LIMITED
Address:	South Gouripur, Ashulia, Savar Ashulia, Savar Dhaka Bangladesh
Country:	Bangladesh
Province:	Dhaka
City:	Ashulia, Savar
Zip Code:	
Audit Duration:	2 Days
Re-Audit:	Re-Audit After 0 Months
Draft Report Date :	30 May 2016
Final Report Date :	24 June 2016
Are all Action Items From Previous Assessment Completed?:	N/A
Buildings in Complex :	7 nos; 1) 1 storied steel shed; 2) 2 storied steel shed; 3) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room); 4) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance); 5) Utility Area (Boiler, Compressor, Substation and Transformer); 6) Generator room; 7) Proposed pump house.
Number of Building Levels (Stories) :	1) 1 storied steel shed, Story above grade.; 2) 2 storied steel shed, Story above grade;
Approximate Building Area (SF) :	1) 1 storied steel shed about 28,416 sft; 2) 2 storied steel shed about 35,842 sft.
Date of Building Construction :	1) 1 storied steel shed - 2010; 2) 2 storied steel shed - 2015; 3) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room) - 2010; 4) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance) - 2010; 5) Utility Area (Boiler, Compressor, Substation and Transformer) - 2015; 6) Generator room- 2015; 7) Proposed pump house - 2015.
Date of Last Building Renovation/Addition :	No renovation or addition has been noticed.
Is the Building mixed use?:	No
Ancillary Structures in Complex :	5 nos; 1) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room); 2) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance); 3) Utility Area (Boiler, Compressor,

	Substation and Transformer); 4) Generator room; 5) Proposed pump house.
Number of Ancillary Levels (Stories) :	1) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room), Stories above grade; 2) 3 storied RCC building, Stories above grade (Canteen, Accessories store, Fabrics store & Maintenance); 3) 1 storied Utility Area, Storey above grade (Boiler, Compressor, Substation and Transformer); 4) 1 storied Generator room, Storey above grade; 5) 1 storied Proposed pump house, Storey above grade.
Approximate Ancillary Structures Area (SF) :	1) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room) about 16,700 sft; 2) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance) about 12,400 sft; 3) Utility Area (Boiler, Compressor, Substation and Transformer) 1,110 sft; 4) Generator room about 200 sft; 5) Proposed pump house about 105 sft.
Number of Occupants :	1067; 1) 1 storied steel shed-385; 2) 2 storied steel shed -610; 3) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room) - 45; 4) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance) - 23; 5) Utility Area (Boiler, Compressor, Substation and Transformer) – 4.
Exterior Facade Description :	1) 1 storied steel shed: Steel frame work filled with 5 inch thick ceramic brick wall with window made of aluminum frame with glass. 2) 2 storied steel shed: Steel frame work filled with 5 inch thick plastered masonry brick wall with window made of aluminum frame with glass.. 3) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room): RCC frame work filled with 5 inch thick plastered masonry brick wall with window made of aluminum frame with glass. 4) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance): RCC frame work filled with 5 inch thick plastered masonry brick wall with window made of aluminum frame with glass. 5) Utility Area (Boiler, Compressor, Substation and Transformer): RCC frame work filled with 5 inch thick plastered masonry brick wall with window made of aluminum frame with glass. 6) Generator room: Steel frame work filled with 5 inch thick masonry brick wall with window in addition to MS grill and louver. 7) Proposed pump house: RCC frame work filled with 5 inch thick masonry brick wall.
Structural System Description :	1) 1 storied steel shed: Frame work of the building is composed of zinc aluminum corrugated metal sheet, structural steel beams and structural steel columns supported by RC spread footing. 2) 2 storied steel shed: Frame work of the building is composed of zinc aluminum corrugated metal sheet, structural steel beams and structural steel columns supported by RC spread footing. 3) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room): Rc slabs, beams and columns supported by isolated spread footings. 4) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance): Rc slabs, beams and columns supported by isolated spread footings. 5) Utility Area (Boiler, Compressor, Substation and Transformer): Rc slabs, beams and columns. No record of footing is available. 6) Generator room: Frame work of the building is composed of zinc aluminum corrugated metal sheet, structural steel Angle and MS pipe columns. No record of footing is available. 7) Proposed pump house: Rc slabs, beams and columns supported by isolated spread footings. No record of footing is available.
Issues were not found during the structural integrity assessment that required the Emergency Escalation Protocol (and referral to NTC Review Panel)?:	Yes

ASSESSMENT FINDINGS

Structural System Design		
Question:	Are the available FoS for the columns adequate based on Preliminary calculation?	
Priority Level:	High	
Non-Compliance Level:	2	
Description:	Preliminary column FoS calculated values of 2 storied steel shed and 4 storied RCC building based on as-built configuration confirmed via structural drawings have been found adequate. 3 storied RCC building has been found inadequate.	
Source of Findings:	Uploaded Document: Document attached	
Suggested Plan of Action:	Some doubts about safety; Concrete strength shall be confirmed by core cutting. Under guidance from a qualified structural engineer arrange Detail Engineering Assessment of the structure.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Provide results of preliminary calculations in space provided. a) column capacity; FoS > 1.86 - Safe b) column capacity; FoS 1.5 -1.86 - Needs Evaluation c) Column capacity; FoS 1.25-1.5 - Needs Evaluation d) Column capacity; FoS <1.25 - Unsafe In case of a critically low FoS (<1.25), consider Immediate Escalation Protocol	
Question:	Are credible structural design documents available for review and kept on site?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	1) 1 storied steel shed: Credible structural design documents are available for review and kept on site. 2) 2 storied steel shed: Credible structural design documents are available for review and kept on site. 3) 4 storied RCC building (Office, Child care, Medical, Sample & Fire control room): Credible structural design documents are available for review and kept on site. There is a Self supported stair but design documents are not available. 4) 3 storied RCC building (Canteen, Accessories store, Fabrics store & Maintenance): Structural design documents are available but do not match with existing structure and therefore shall be consider not credible. 5) Utility Area (Boiler, Compressor, Substation and Transformer): Undocumented structure. 6) Generator room: Undocumented structure. 7) Proposed pump house: Undocumented structure.	
Source of Findings:	Document Review: Seen all document	
Suggested Plan of Action:	A qualified structural engineer shall be engaged to prepare credible as-built documents for 3 Storied building; Utility Area (Boiler, Compressor, Substation and Transformer); Generator room and Proposed pump house based on the	

	requirements of Part 8 Section 8.19 of the Alliance Standard.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.19 Required Structural Documentation for New and Existing Factories	
Question:	Have provisions been made in floors or decks for a concentrated load (such as heavy equipment, water tanks, stored materials, etc) applied at a location wherever this load acting upon an otherwise unloaded floor would produce stresses greater than those caused by a uniform load?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	1- Plastic water tank has been found over 3 storied building. 2- Plastic water tanks have been found over 4 storied building.	
Source of Findings:	Photograph: Picture attached	
Suggested Plan of Action:	A qualified structural engineer shall be engage to confirm and document that provisions have been made to accommodate concentrated loads. If provisions have not been made, have a qualified structural engineer develop a remediation plan.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.13 and 8.14	
Question:	Are Certificates of Occupancy available for review?	
Priority Level:	Low	
Non-Compliance Level:	2	
Description:	Certificate of Occupancy was not available for review.	
Source of Findings:	Document Review: Seen all document	
Suggested Plan of Action:	Certificate of occupancy shall be obtained from appropriate authority.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.3 Preliminary Structural Assessment	
Structural System Construction		

Question:	Have all areas of needed maintenance, including areas with efflorescence, dampness, standing water on rooftops, and corrosion been addressed.	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	Signs of standing water have been noticed over roof of 3 storied RCC building and utility building. Corrosion has been noticed in north-east side structural steel column of 1 storied steel shed.	
Source of Findings:	Photograph: Standing water	
Suggested Plan of Action:	Under guidance from a qualified structural engineer, address all areas of needed maintenance by correcting the identified issues.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.26 Durability and Maintenance	
Question:	If yes, have the structural members constructed with MCAC been investigated by an appropriate program of in-situ testing and representative destructive testing or core samples?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	No testing was done for assessment of the strength of the concrete. FoS have been found less than 1.86 in 3 storied RCC building.	
Source of Findings:	Uploaded Document: FoS calcs uploaded under column FoS question.	
Suggested Plan of Action:	Concrete strength shall be confirmed by core cutting in 3 storied RCC building.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Reference Alliance Standards Part 7 Building Materials Section 7.2 Masonry-chip aggregate concrete (MCAC)	
Question:	Are any structural elements constructed with MCAC exposed to rainfall or other sources of water sealed with a protective coating to prevent water intrusion?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	4 storied RCC building roof has been found without protective coating.	
Source of Findings:	Visual Assessment: Visual inspection	
Suggested Plan of	Provide a protective coating at the structural elements constructed with MCAC	

Action:	exposed to rainfall or other sources of water. Have protective coating approved by the Alliance or a qualified structural engineer.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standards Part 7 Building Materials Section 7.2 Masonry-chip aggregate concrete (MCAC).	
Question:	Are structural steel members free of corrosion, physical damage or other types of deterioration?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	Corrosion has been noticed in north-east side structural steel column of 1 storied steel shed.	
Source of Findings:	Photograph: Steel corrosion	
Suggested Plan of Action:	Complete further testing on areas of deterioration and have a qualified structural engineer develop a remediation plan.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.26	
Question:	Are all non-structural elements suspended from, attached to, or resting atop the structure adequately anchored and braced to resist earthquake forces?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	Hanger bar used for running electric cables of production shed and plastic tank over roof of 4 storied RCC building and 3 storied RCC building have not been anchored and braced adequately to resist earthquake force.	
Source of Findings:	Visual Assessment: Unanchored items noted during review.	
Suggested Plan of Action:	Adequately anchor and brace all non-structural elements to resist earthquake forces to comply with the BNBC and Alliance Standard.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standards Part 8 Section 8.18 Seismic Bracing of Key Non-Structural Elements and 2006 BNBC Part 6	

Question:	Is the building free of active signs of water intrusion or ponding due to lack of performance of the façade system?	
Priority Level:	Low	
Non-Compliance Level:	1	
Description:	All buildings except 1 storied steel shed have been found free of active signs of water intrusion due to lack of performance of the façade system.	
Source of Findings:	Photograph: Picture attached	
Suggested Plan of Action:	Repair the exterior façade system to prevent water intrusion.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.26 Durability and Maintenance	
Structural Safety Programs		
Question:	Is a program in place to ensure that the live loads for which a floor or roof is or has been designed will not be exceeded?	
Priority Level:	Medium	
Non-Compliance Level:	2	
Description:	There is no such program in place (floor and roof) ensuring that the live loads for which a floor or roof is or has been designed will not be exceeded.	
Source of Findings:	Document Review: No documented load management plan.	
Suggested Plan of Action:	Develop a program to ensure that all live loads for which a floor or roof has been designed for will not be exceeded. The designated Load Manager shall oversee this program and ensure it is enforced.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 13 Section 13.7 and Part 8 Section 8.9.	
Question:	Have Load Plans been prepared for each floor documenting the actual maximum operational loading that is intended and/or allowable on each floor.	
Priority Level:	Low	
Non-Compliance Level:	2	
Description:	Load plans have not been prepared, documenting the actual maximum operational loading.	
Source of Findings:	Document Review: Documents review on-site.	
Suggested Plan of	Have a qualified structural engineer develop Floor Loading Plans per the	

Action:	requirements of Part 8 Section 8.20.5.3.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.10 Floor Loading Plans (Load Plans)	
Question:	Are Floor Load Plans posted as required?	
Priority Level:	Low	
Non-Compliance Level:	2	
Description:	Load plans have not been prepared.	
Source of Findings:	Document Review: Documents reviewed on-site.	
Suggested Plan of Action:	Have a qualified structural engineer prepare load plans including the information required in Section 8.20 of the Alliance Standard. Floor load plans should be visibly posted on all levels of all buildings.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.20.5.3	
Question:	Are areas used for storage of work materials and work products, clearly marked to indicate the acceptable loading limits as described in the Load Plan for that floor?	
Priority Level:	Low	
Non-Compliance Level:	2	
Description:	Areas used for storage of work materials and work products have not been marked indicating the acceptable loading limits.	
Source of Findings:	Visual Assessment: No visible markings.	
Suggested Plan of Action:	Provide signage or the appropriate markings at all areas used for storage to indicate the acceptable loading limits detailed in the Load Plan.	
Suggested Deadline Date:	15 Aug 2016	
Standard:	Alliance Standard Part 8 Section 8.11 Floor Load Markings	
Question:	Is a designated representative (Factory Load Manager), who is onsite full time, trained regarding the structural floor capacity, and serves as an ongoing vendor resource and monitor of operational factory floor loadings?	
Priority Level:	Low	
Non-Compliance Level:	2	
Description:	No designated factory load manager was noted during the assessment.	

Source of Findings:	Document Review: No such documentation.
Suggested Plan of Action:	Designate a representative as the Factory Load Manager. The Factory Owner shall ensure that at least one individual, the Factory Load Manager who is located onsite full time at the factory, is trained in calculating operational load characteristics of the specific factory. The Factory Load Manager shall serve as an ongoing resource to RMG vendors and be responsible to ensure that the factory operational loads do not at any time exceed the factory floor loading limits as described on the Floor Loading Plans.
Suggested Deadline Date:	15 Aug 2016
Standard:	Alliance Standards Part 8 Section 8.9 Factory Load Manager